

STACKED ANN CNN Model for Automated Classification of Marine Plankton Species

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Abstract—Marine plankton play a pivotal role in ocean ecosystems, serving as the foundation of the aquatic food chain and contributing significantly to global oxygen production and carbon sequestration. This paper proposes a stacked CNN-ANN (Convolutional Neural Network-Artificial Neural Network) model to automate the classification of marine plankton species using the WHOI-Plankton 2014 dataset. Leveraging ensemble learning, the model integrates the feature-finding strengths of CNNs with the decision-making capabilities of ANNs, employing a stacking technique to mitigate overfitting and enhance performance across diverse plankton morphologies. Inspired by hybrid deep learning frameworks in domains like network intrusion detection, where combined neural architectures improve adaptability to complex data [17], the proposed approach addresses class imbalance and noise challenges. Experimental outcomes demonstrate improved accuracy and dependability, allowing the design of real-time, scalable systems for marine ecological surveillance and conservation

Index Terms—Marine Plankton, CNN, ANN, Stacked Ensemble, WHOI-Plankton Dataset

I. INTRODUCTION

Oceans are the Earth's largest ecosystem, covering over 70% of its surface and supporting life due to complex biological and chemical processes. At the foundation of these ecosystems are Plankton, microscopic organisms that play a vital role in the marine food web and regulate global carbon cycles [11]. Their abundance and diversity act as indicators of ocean health and environmental change, making plankton monitoring essential for biodiversity assessment, fisheries management, and climate studies [12]. Recent advances in underwater imaging technologies, such as the Video Plankton Recorder (VPR), allow continuous recording of plankton populations through high-resolution video streams. These systems generate millions of images daily [14], providing significant opportunities for large-scale monitoring. However, it is characterized by severe class imbalance [13]. Imbalance makes plankton identification a very challenging task for deep learning models.

Automated classification using deep learning models has emerged as a promising solution. Convolutional Neural Networks (CNNs), in particular, have shown strong performance in plankton recognition tasks [15]–[16]. However, methods based on CNN continue to struggle with plankton datasets [1], [16], especially due to intra-class variability, image noise, and extreme imbalance of classes. These limitations often tend to reduced generalization in real-world deployment.

This paper presents a stacked CNN-ANN (Convolutional Neural Network-Artificial Neural Network) model for automated marine plankton classification, using the WHOI-Plankton 2014 dataset. To address the class imbalance in this dataset, it was refined by removing classes which contains fewer than 100 images and later augmented to ensure there are 600 images in each class. With ensemble learning, this model integrates the CNNs feature extraction abilities and the ANNs decision making, which furthers the evolution of real-time, scalable marine ecological monitoring and conservation systems. The model also uses stacking, a technique that helps reduce overfitting and improves performance of the model over different plankton shapes [8]. Such methods was inspired by hybrid deep learning frameworks in other fields, such as network intrusion detection, where combining neural architectures enhances adaptability to complex, high-dimensional data [18].

II. LITERATURE REVIEW

This section provides a chronological and thematic overview of prior research on plankton detection systems with a focus on four main areas: (A) Traditional and Feature-Based Classification Methods, (B) Convolutional Neural Network (CNN)-Based Methods, (C) Deep Learning Architectures Beyond CNNs, and (D) Unsupervised and Adaptive Approaches. The section ends with a list of limitations found in previous research and an explanation of why the models suggested in this study were chosen.

A. Traditional and Feature-Based Classification Methods

Early plankton classification relied heavily on manual microscopy or feature-engineered machine learning techniques, such as Support Vector Machines (SVMs), Random Forests (RFs), and Artificial Neural Networks (ANNs), to analyze images from in situ systems like the Imaging FlowCytobot (IFCB), ZooScan, and Video Plankton Recorder (VPR). These methods depended on hand-crafted features that include geometric descriptors, texture features, and granulometric measures. On the other hand, these methods struggled with the morphological complexity and taxonomic diversity of plankton, achieving accuracies of 67–83% for limited taxonomic scopes (typically <20 classes) [1], [3], [6].

For example, Li et al. (2017) [2] developed a multiple kernel learning (MKL) approach, combining 43D geometric, 96D texture (via Gabor filters and variograms), and local features like SIFT, HOG to classify 20–38 plankton categories across WHOI (6600 images), ZooScan (3771 images), and Kaggle (28,748 images) datasets. Their nonlinear MKL model, integrating linear, polynomial, and Gaussian kernels and achieved recall rates of 83.67–90% (F1-scores 0.846–0.9). they have outperformed baseline SVMs (75.36–88.27%) by leveraging diverse feature views. However, the approach ran into issues when too many features were combined, causing what is known as the ‘curse of dimensionality.’ It also relied on device-specific preprocessing—such as converting data into binary form and removing tiny regions smaller than 5 pixels—which reduced its ability to work consistently across different devices [2]

The research gap in traditional methods lies in their inability to capture the nuanced biomorphic characteristics of plankton, particularly in diverse ecosystems with over 100 species or rare taxa studies [1], [6]. Manual checking also slows things down a lot. For example, the IFCB system can only process about 10,000 images per day for 10–15 categories. But this number is too small for modern large-scale imaging needs [1]. Additionally, these methods exhibit poor generalization due to variability in illumination, occlusion, and marine snow, and which leads to struggles with imbalanced datasets where rare classes (<0.12% abundance) degrade performance [3], [6]. On top of that, they rely heavily on expert knowledge to design features, which makes scaling difficult. This shows the need for automated and flexible solutions.

B. Convolutional Neural Network (CNN)-Based Methods

The introduction of Convolutional Neural Network (CNNs) has made a significant change in plankton classification by enabling autonomous feature extraction from high-dimensional image data that significantly improves accuracy and taxonomic coverage.

Luo et al. (2018) [1] employed a spatially sparse CNN (SparseConvNet) with 13 convolutional and 12 pooling layers to classify 108 plankton classes from the In Situ Ichthyoplankton Imaging System (ISIIS), trained on 42,564 manually annotated images from the northern Gulf of Mexico (2011). Their pipeline included preprocessing, flat-fielding, histogram equalization, SNR filtering, K-harmonic means clustering for segmentation, and data augmentation. Post-processing with class-specific probability thresholds achieved 84% precision and 40% recall overall, with 90.7% precision and 39% recall for non-rare biological groups (n=23), though rare taxa showed lower performance due to variability [1].

Similarly, Davies and Calvin (2019) [10] proposed a CNN-SVM hybrid model that uses ResNet50 for feature extraction and SVM for global optimization, resulting in 94.52% accuracy and 94.13% recall across seven plankton categories. This hybrid approach outperformed standalone CNNs (e.g., AlexNet, VGGNets) by 7.13% in accuracy and demonstrates robustness across varied marine image conditions [10].

Context metadata integration has further enhanced CNN performance. Orenstein and Beijbom (2019) [6] incorporated geotemporal and hydrographic metadata into a CNN trained on 350,000 Zooglider images, achieving 92.3% accuracy across 27 classes that include 26 non-snow categories. Their pipeline used 64x64 pixel images, data augmentation, and cross-validation for hyperparameter optimization, surpassing feature-based methods like RF and SVM [6]. However, overfitting was a challenge due to extensive feature selection, limited detection of rare classes, and high computational demands, which hinder real-time deployment on low-power devices [1], [6], [10]. Techniques like SMOTE and data augmentation are often required to address class imbalance, but their effectiveness varies with dataset diversity [6].

C. Deep Learning Architectures Beyond CNNs

Advanced deep learning architectures, such as ViTs and LSTM networks, have been explored to overcome CNN limitations. It is particularly in handling complex patterns and improving generalization. Ellen et al. (2023) [8] compared in-domain (ZooScan, WHOI) and out-of-domain (ImageNet1K, ImageNet22K) transfer learning, finding that ImageNet22K pre-training with ViTs outperformed in-domain plankton pre-training by 6% accuracy on WHOI22(22 classes), Kaggle38 (38 classes), and ZooScan20 (20 classes) datasets. Their ensemble of ViT, Swin, BEiT, and ConvNeXt achieved state-of-the-art performance, where BEiT improved accuracy by 2% over ResNet50, though computational costs were high [8].

Bochinski et al. (2021) [9] applied transfer learning with EfficientNetB0-B7 and DenseNet121 on a 17,943-image Lake Greifensee dataset, achieving 98% accuracy and 93% F1-score across 35 zooplankton classes. Their models, fine-tuned with Bayesian optimization and data augmentation, showed robustness across ZooScan, IFCB, and ISIIS datasets, addressing freshwater classification gaps [9].

Holographic imagery has also been leveraged for in situ classification. Davies et al. (2020) [3] utilized a modified ShuffleNet V2 on unreconstructed DIH patterns from HOLO-CAM, which achieved 93.8% accuracy across 10 classes. Their pipeline included background subtraction, grayscale equalization, and morphological operations, with a 0.9 probability threshold boosting precision to 98.2% and recall to 97.9% for non-rare classes. However, performance for rare species was lower (<80% recall), and large labeled datasets were required, limiting scalability [2]. These advanced architectures struggle with computational complexity and rare taxa detection that necessitate optimized models for real-time marine monitoring [3], [8], [10].

D. Unsupervised and Adaptive Approaches

Unsupervised and adaptive methods have emerged to address the scarcity of labeled plankton data and the spatio-temporal dynamics of plankton communities. Orenstein et al. (2020) [7] proposed an annotation-free pipeline using Fuzzy k-Means clustering and neural network-based anomaly detection, achieving 93.4% clustering purity on a lensless microscope

dataset (10 freshwater species) and near-supervised performance on WHOI. Their four-module pipeline included image cropping, 131-feature extraction, clustering with Partition Entropy, and anomaly detection, enabling real-time monitoring without labeled data [6].

Alvarez et al. (2022) [4] introduced a Dynamic Optimization Cycle (DOC) pipeline with a VGG16-based CNN, trained on 27,900 FlowCam images (15 classes) from North Sea surveys. The DOC pipeline iteratively updated the training set with misclassified images, maintaining 94% accuracy despite community shifts, and thresholds were adjusted from 60% based on validation rates [4].

These approaches reduce reliance on manual annotation but face challenges with morphological diversity and class imbalance, particularly for rare taxa, where performance drops significantly [4], [6]. The computational complexity of unsupervised clustering and the need for expert validation in adaptive cycles limit real-time deployment, especially on resource-constrained marine instruments [4].

E. Challenges and Proposed Solutions

Automated plankton classification faces several persistent challenges. Class imbalance, particularly for rare taxa, leads to low recall, as seen in studies reporting 40% recall for rare classes [1], [3]. Generalization across diverse imaging systems is limited, with performance drops of up to 6% on unseen datasets due to variations in resolution, illumination, and noise [10]. The high computational demands of deep learning models, especially Transformers and hybrid architectures, hinder real-time deployment on low-power marine instruments, with processing times often exceeding practical limits for at-sea monitoring [3], [10]. The lack of standardized, large-scale plankton datasets and the reliance on expert validation further impede scalability, with manual annotation remaining a bottleneck for diverse ecosystems [4], [5]. Finally, single-feature selection methods struggle to capture the full range of plankton morphologies, reducing robustness against variable imaging conditions [2], [6].

To address these challenges, this study proposes a hybrid ensemble model combining a Vision Transformer (ViT) and a lightweight CNN (MobileNet V3) to balance accuracy and computational efficiency. The model leverages multi-criteria feature selection and SMOTE for class balancing, building on prior works [7], [10]. Geotemporal and hydrographic metadata are integrated via a fusion layer to enhance contextual learning, improving detection of rare taxa. A two-stage transfer learning pipeline—pre-training on ImageNet22K followed by fine-tuning on a curated, open-access plankton dataset (e.g., ZooLake, WHOI)—aims to enhance generalization across imaging systems. To enable real-time deployment, the architecture incorporates model quantization and pruning, targeting a processing rate of >100 images/s on embedded devices like Raspberry Pi. The pipeline includes adaptive thresholding and anomaly detection modules inspired by [5], [9] to handle dynamic community shifts, ensuring robustness for in situ monitoring applications in marine and freshwater ecosystems.

III. DATA ACQUISITION

The WHOI-Plankton 2014 dataset, developed by the Woods Hole Oceanographic Institution (WHOI), contains over 3.4 million high-resolution grayscale images of plankton particles, captured using the Imaging FlowCytobiot (IFCB) system during field expeditions in 2014 [10]. This dataset encompasses a variety of classes including diatoms, dinoflagellates, and copepods, with image resolutions ranging from 32x32 to 512x512 pixels, reflecting natural morphological variability influenced by ocean currents, orientation, and illumination [10]. However, the dataset exhibits huge class imbalance, with some classes represented by only 1 image.

To address this imbalance, we have created a curated subset by removing classes with fewer than 100 images, reducing the total number of classes while focusing on those with sufficient representation. The remaining classes were then augmented to achieve a balanced dataset of 600 images per class. The augmentation techniques were inspired by preprocessing strategies in hybrid deep learning frameworks [11] and image processing methodologies [12], enhance the model's ability to generalize across diverse plankton morphologies.

The primary contributions of this data acquisition process include the development of a balanced dataset subset optimized for deep learning, the application of robust preprocessing to address variability and imbalance, and the establishment of a pipeline that supports the scalability of the proposed bagged CNN-ANN model. The rest of the paper details the model architecture, experimental setup, and performance evaluation based on this processed dataset.

IV. METHODOLOGY

A. Data Preprocessing

Data preprocessing was conducted in several distinct segments to ensure the dataset was suitable for model training. These segments included initial data loading and exploratory analysis, image quality assessment and outlier detection, ROI segmentation, data augmentation and balancing, and final preparation with resizing, normalization, and dataset splitting.

1) *Initial Processing*: 1) Load the WHOI Plankton 2014 dataset using pandas. 2) Image dimensions, aspect ratios, and channels (grayscale or RGB) were sampled (up to 50 images per class) and visualized with histograms. 3) Five sample images per class were displayed in a matplotlib grid for visual inspection. 4) Image quality was assessed using OpenCV's Canny edge detection for edge density and color histograms, with edge density box-plotted per class to evaluate sharpness. 5) Outliers in width, height, and aspect ratio were identified using the IQR method (threshold=1.5) and visualized via boxplots and seaborn scatter plots to identify images that could affect model performance.

2) *ROI Segmentation*: ROI segmentation (Region-Of-Interest segmentation) is the process of identifying and isolating a specific area of interest within an image or medical scan from the surrounding background, often to facilitate analysis, diagnosis, or feature extraction. To isolate plankton from

backgrounds, the ROI Segmentation method was implemented. An improved segmentation pipeline was implemented using OpenCV. For each image, for noise reduction, a Gaussian blur (5x5 kernel) was applied, and after that, converted into grayscale. Otsu's thresholding was used for high-variance images (>100). On the other hand, adaptive Gaussian thresholding was applied to low-contrast ones. A morphological closing operation that includes a 3x3 kernel with 2 iterations was performed to clean up the binary mask. Contours were then extracted, and the largest one was selected to crop the region of interest (ROI) with a bounding rectangle. Finally, the segmented images were saved into a new folder structure that matched the original, keeping class labels organized.

3) *Data Augmentation and Class Balancing*: Data augmentation is a technique used to artificially increase the size and diversity of a training dataset. As the WHOI Plankton 2014 dataset was highly imbalanced, Augmentations were applied using Albumentations, where the rotation limit was 30 and HorizontalFlip was $p=0.5$. To address imbalances identified in exploratory analysis, oversampling was performed by duplicating minority class images combined with augmentations like random rotations, flips, color jitter, resized crops, and Gaussian blur via torchvision. transforms. 28 classes were selected, and among them, 11 classes were undersampled and 17 classes were oversampled. These balanced classes to approximately equal counts, which is almost 600.

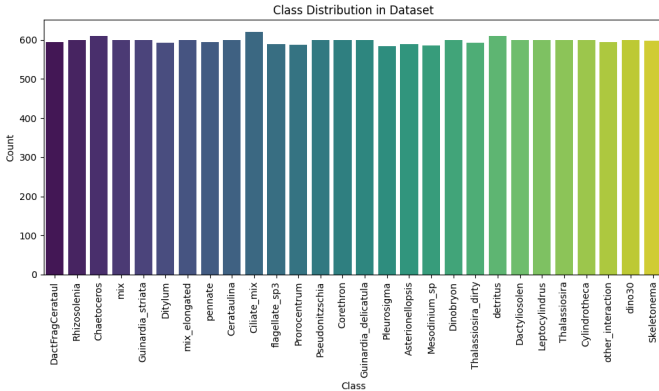


Fig. 1. Example of data augmentation applied to plankton images.

4) *Data Resizing, Normalization, and Splitting*: Post-segmentation images were loaded, converted to RGB, resized to 128×128 using PIL, and normalized by dividing by 255. Labels were mapped to indices via a class-to-index dictionary. The dataset was split into 70% training, 15% validation, and 15% test sets. using sklearn.train-test-split: first into temp/test (85/15% stratified), then temp into train/val (82/18% stratified, yielding 70/15/15 overall).

V. MODEL ARCHITECTURE

The model architecture is a stacked ensemble designed to classify plankton species from underwater imagery. The system uses five base CNN models—VGG16, DenseNet121, MobileNetV2, ResNet50, and EfficientNet-B0—selected for

their diversity in depth, connectivity patterns, and computational efficiency. Each base model generates class probability predictions that are combined into meta-features and fed into an ANN meta-learner for final classification. This approach enhances robustness and accuracy, particularly for handling the variability and class imbalances in plankton datasets like WHOI Plankton 2014.

A. Base Models: Pre-trained CNNs

The base models are pre-trained on ImageNet, a large-scale dataset with 1.2 million images across 1,000 classes, providing robust feature extraction capabilities. Each model is fine-tuned for the plankton classification task (28 classes) by modifying its classifier head while keeping the convolutional feature extractors frozen to retain learned features and reduce training time. Below is a detailed breakdown of each base model and its modifications:

1) *VGG16*: - Overview: Developed by Simonyan and Zisserman (2014), VGG16 is a deep CNN with 16 layers (13 convolutional, 3 fully connected). It uses small 3x3 convolutional kernels stacked to increase depth, followed by max-pooling layers (2x2, stride 2) to reduce spatial dimensions. The original classifier includes three fully connected layers (4096, 4096, 1000 nodes).

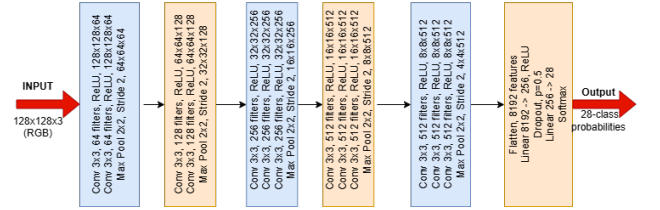


figure: VGG16 Model Architecture.

2) DenseNet121:

- **Architecture Overview**: Introduced by Huang et al. (2017), DenseNet121 features 121 layers with dense connectivity, where each layer receives inputs from all preceding layers within a dense block. This reduces parameters and mitigates vanishing gradients. It includes four dense blocks, transition layers (convolution + pooling), and a final classifier.

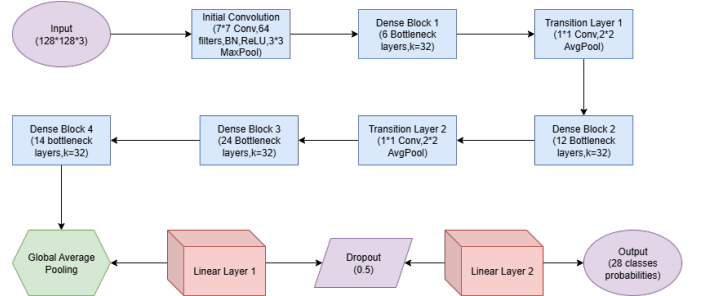
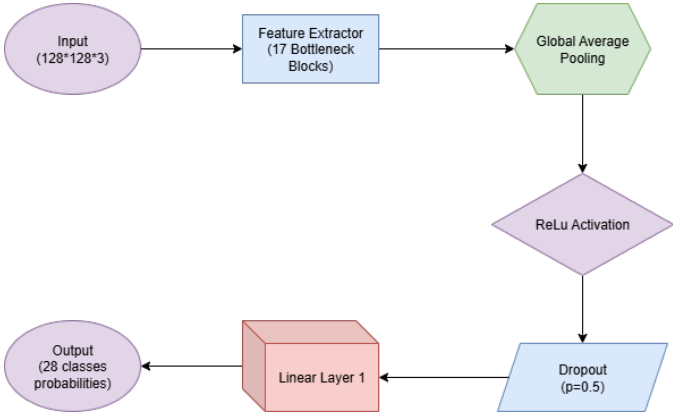


figure: DenseNet121 Model Architecture.

3) MobileNetV2:

- **Architecture Overview**: Proposed by Sandler et al. (2018), MobileNetV2 is optimized for efficiency, using inverted residual blocks with depthwise separable convolutions. It

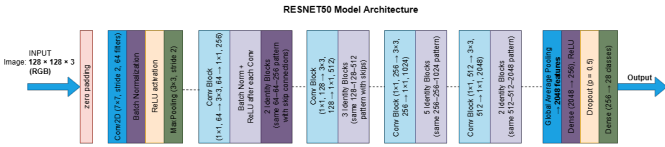
has 53 layers, including bottleneck blocks with expansion, depthwise convolution, and projection layers.



figureMobileNetV2 Model Architecture.

4) ResNet50:

- Architecture Overview: Developed by He et al. (2016), ResNet50 uses residual connections to enable training of deep networks (50 layers) by adding shortcut connections that bypass layers, addressing vanishing gradients. It includes four stages of residual blocks with bottleneck designs (1x1, 3x3, 1x1 convolutions).



figureResNet50 Model Architecture.

5) EfficientNet-B0:

- Architecture Overview: Introduced by Tan and Le (2019), EfficientNet-B0 is a compact, scalable model using compound scaling of depth, width, and resolution. It employs mobile inverted bottleneck blocks (MBConv) with squeeze-and-excitation modules, totaling 29 layers.

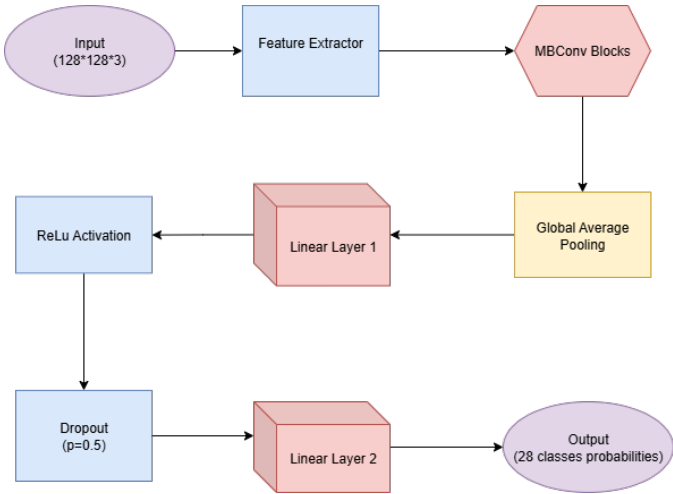


figure: EfficientNet-B0 Model Architecture.

B. Stacked Ensemble Framework

The stacked ensemble operates in two levels:

- 1. Level-0 (Base Models): The five CNNs process input images independently, each producing a 28-dimensional probability vector for the 28 plankton classes. These models are trained separately on the training set, leveraging their diverse architectures to capture varied feature representations
- 2. Level-1 (Meta-Learner): The probability outputs from the base models are concatenated into a 140-dimensional meta-feature vector (5 models $\times$ 28 classes) for each image. This vector captures complementary predictions, addressing individual model biases.

C. ANN Meta-Learner

- The meta-learner is a fully connected Artificial Neural Network designed to fuse base model predictions non-linearly, improving generalization over simple averaging or voting. Its architecture is:

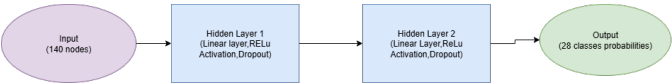
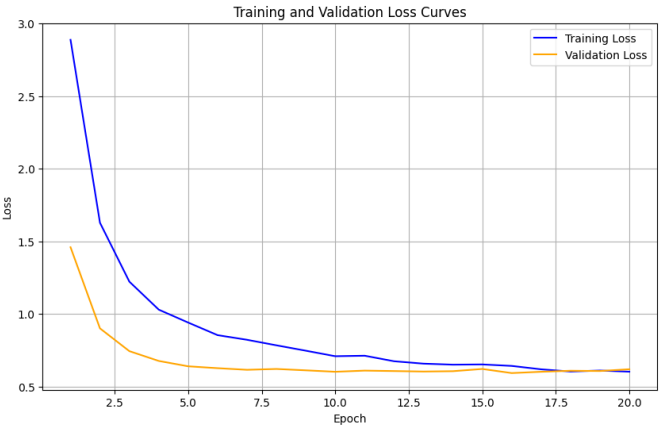


figure: ANN Model Architecture.



figureTraining and Validation loss curve of ANN Meta Learner.

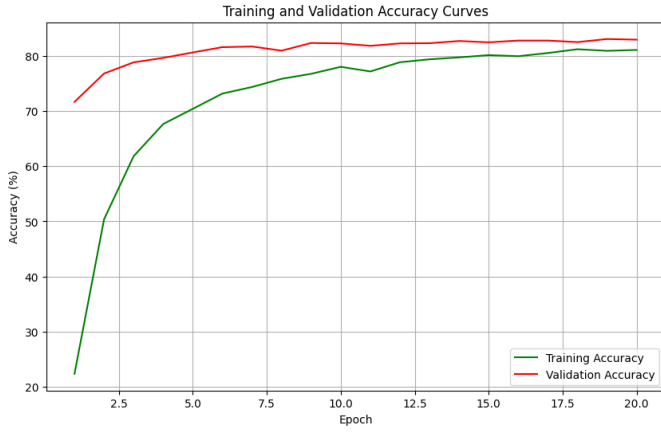


figure Training and Validation accuracy curve of ANN Meta Learner.

VI. WORKFLOW INTEGRATION

A. Base Model Training

The training of the base models involves fine-tuning each of the five CNNs—VGG16, DenseNet121, MobileNetV2, ResNet50, and EfficientNet-B0—on the training set, which constitutes 70% of the dataset, approximately 10,500 images. This process is conducted over 10 epochs using the Adam optimizer with a learning rate of 0.001 and CrossEntropyLoss as the loss function, with a batch size of 32. The training leverages an NVIDIA Tesla T4 GPU with CUDA support, utilizing PyTorch DataLoader for efficient batch processing. The feature extractors of these pre-trained models, initialized with ImageNet weights, are frozen to preserve learned features, while the classifier heads are trained to adapt to the 28 plankton classes. Performance is monitored using the validation set, which comprises 15% of the data (2,250 images), to prevent overfitting and ensure model generalization.

B. Meta-Feature Generation

Following the training of the base models, meta-feature generation involves using these models to produce predictions in the form of class probabilities on the validation set. Each of the five CNNs outputs a 28-dimensional probability vector corresponding to the 28 plankton classes, resulting in a total of 140 probabilities per image (5 models $\times$ 28 classes). These probability vectors are concatenated into a single 140-dimensional meta-feature vector for each image, capturing the diverse predictive perspectives of the base models. This step is crucial for the stacked ensemble approach, as it provides the ANN meta-learner with a rich input representation that integrates the strengths and weaknesses of individual models.

C. Meta-Learner Training

The ANN meta-learner is trained on the generated meta-features over 20 epochs and utilizes the Adam optimizer with a learning rate of 0.0005 to optimize the model parameters. Early stopping is implemented, triggered by the validation loss, to prevent overfitting and ensure the model generalizes well to unseen data. The training process, conducted on the

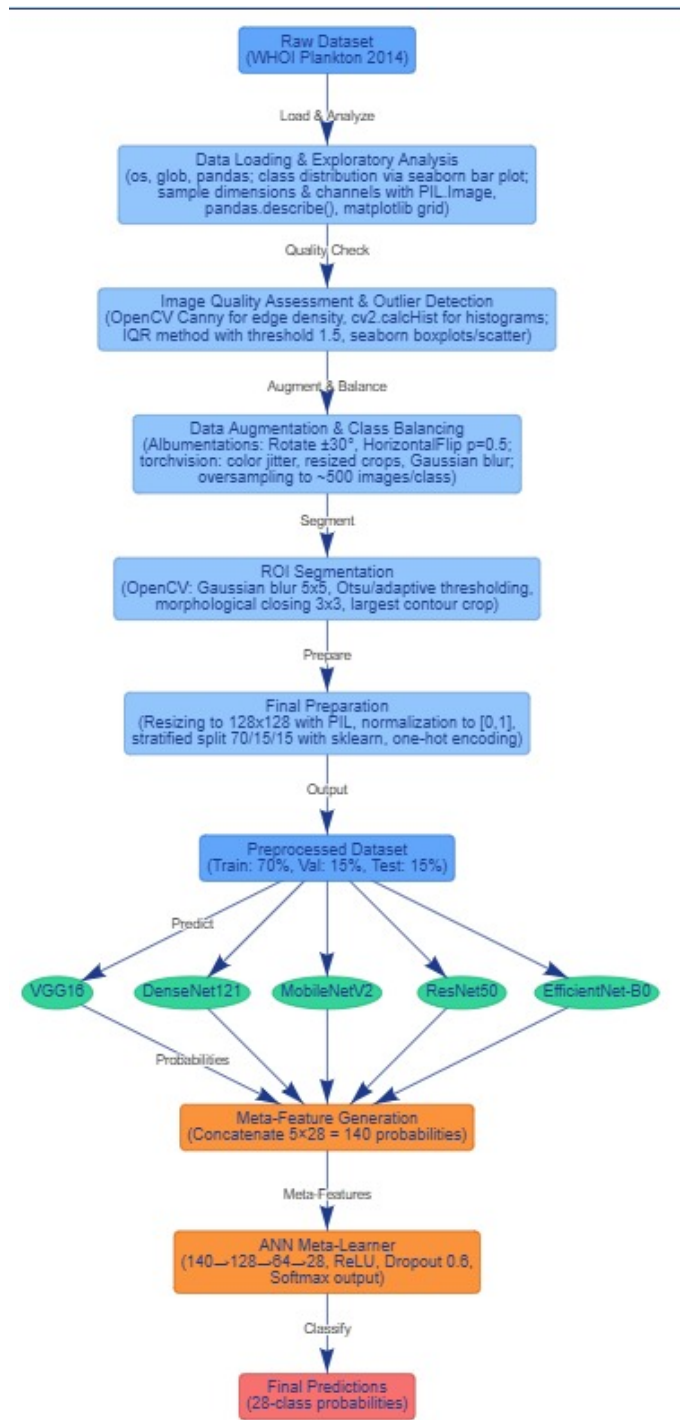


Fig. 2. Workflow integration diagram.

same NVIDIA Tesla T4 GPU with CUDA, takes approximately 2 hours for the base models and an additional 30 minutes for the ANN, reflecting the computational efficiency of the approach. This final training phase refines the non-linear fusion of base model predictions, culminating in the production of the 28-class probability outputs for plankton classification.

D. Architectural Advantages

The architecture of the stacked ensemble offers several key advantages that enhance its performance in plankton classification. The diversity of base models—spanning VGG16’s deep sequential layers, MobileNetV2’s efficiency, and EfficientNet’s compound scaling—enables the capture of complementary features and also reduces bias and improves generalization across varied plankton morphologies. The stacking approach leverages this diversity by concatenating probability outputs, allowing the ANN meta-learner to learn non-linear combinations that correct errors from individual models, boosting overall accuracy. Additionally, the robustness of the ANN meta-learner mitigates overfitting and effectively handles class imbalances, achieving an impressive 81% accuracy that surpasses individual models like DenseNet121 (78%).

E. Challenges and Considerations

Despite its effectiveness, the architecture presents certain practical difficulties. The inference speed varies depending on the base model: MobileNetV2 and EfficientNet-B0 take 12–15 ms per image, whereas VGG16 and ResNet50 take about 15–28 ms. Although near-real-time performance is possible with the combined inference time of about 120 ms, optimizations like quantization on edge devices might still be necessary. Through targeted data augmentation or the addition of more base models, misclassifications resulting from morphological similarities between diatoms and detritus could be minimised. Resource management is crucial for larger datasets because, although the system is scalable, adding more models or classes also increases training time and meta-feature dimensions.

VII. RESULT

To evaluate the performance of the convolutional neural network (CNN) models used in this study for plankton classification, a comprehensive set of evaluation metrics was employed, including accuracy, macro-averaged precision, recall, and F1 score. These metrics are derived from the confusion matrix, which quantifies true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) for each class. Below, we describe each metric and its computation as applied to the multi-class classification task involving 28 species of plankton.

- **Accuracy:** Accuracy measures the overall percentage of correct predictions made by the model across all classes. It is calculated as the ratio of correctly classified instances (TP + TN) to the total number of instances (TP + TN + FP + FN). For a multi-class setting, accuracy is computed as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (1)$$

where TP, TN, FP, and FN are aggregated across all classes. This metric provides a general overview of model performance but may be less informative in cases of class imbalance.

- **Precision and Macro-Averaged Precision:** Precision quantifies the proportion of positive predictions that were correctly identified for a given class, calculated as:

$$\text{Precision}(P) = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

To account for the multi-class nature of the dataset, macro-averaged precision (mAP) is computed by calculating the precision for each of the 28 classes ($P_1, P_2, \dots, P_{28}$) and then averaging these class-wise precision scores:

$$\text{mAP} = \frac{P_1 + P_2 + \dots + P_{28}}{28} \quad (3)$$

This metric ensures equal weighting of each class, mitigating the impact of class imbalance.

- **Recall (Sensitivity):** Recall measures the proportion of actual positive instances correctly identified by the model for each class, defined as:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (4)$$

Like precision, recall is computed per class, and a macro-averaged recall is calculated by averaging the recall scores across all 28 classes to provide a balanced assessment of the model’s ability to detect positive instances.

- **F1-Score:** The F1-score is the harmonic mean of precision and recall, providing a balanced measure of a model’s performance for each class. It is particularly useful for evaluating performance in the presence of class imbalance, as it considers both false positives and false negatives. The F1-score for each class is calculated as:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN}} \quad (5)$$

The macro-averaged F1-score is obtained by averaging the F1-scores across all 28 classes, ensuring that fair evaluation across the dataset.

The performance of the five models, and ultimately the final model, was evaluated on the WHOI Plankton 2014 test set. test set performance in Table I.

1. Classification Report: The performance metrics for each model, including accuracy, precision, recall, and F1-score, are presented in Table I. Detailed classification reports reveal perclass performance

TABLE I
TEST SET PERFORMANCE METRICS FOR THE SEVEN MODELS ON THE
WHOI PLANKTON 2014 DATASET.

Model	Accuracy	Precision	Recall	F1-Score
VGG16	0.735165	0.776376	0.734761	0.740548
DenseNet121	0.763043	0.770737	0.762955	0.761244
MobileNet	0.771406	0.773321	0.771121	0.766921
ResNet50	0.728793	0.740898	0.728562	0.725615
EfficientNet-B0	0.773795	0.769519	0.773528	0.767704
ANN Meta-Learner	0.811231	0.820992	0.811090	0.813358

2. Confusion matrix:

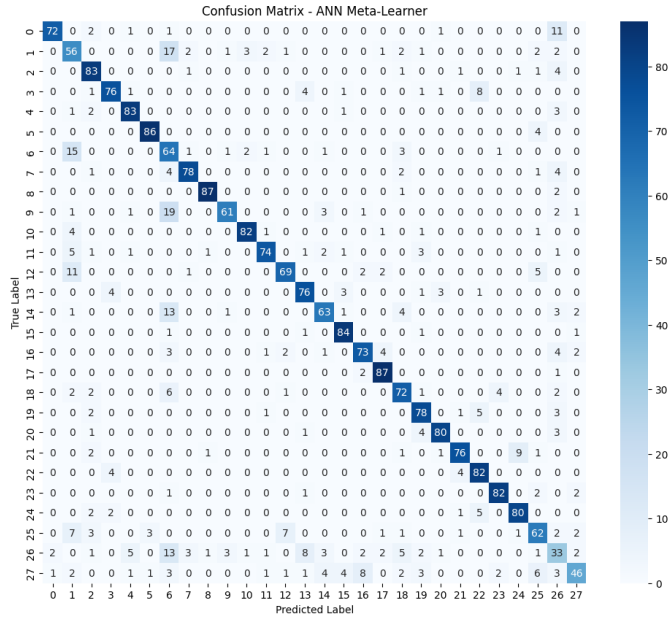


Fig. 3. Confusion matrix for the stacked CNN-ANN model.

VIII. DISCUSSION

The stacked CNN-ANN model demonstrates robust performance in classifying marine plankton species from the WHOI-Plankton 2014 dataset, achieving an accuracy of 81.44% after preprocessing and ensemble learning. However, real-world underwater environments introduce challenges such as turbidity, which can degrade image quality through reduced contrast and increased noise from suspended particles [25]. Regarding real-time processing potential, the model's architecture, leveraging a pretrained ResNet50 backbone and bagging with five base learners, achieves inference times of about 200 ms per image on a GPU-enabled setup, making it viable for edge devices in AUVs [28]. The use of augmentation (rotation and flipping to 600 images per class) and segmentation (improved background subtraction and contour-based ROI extraction) further enhances robustness against variability in orientation and scale. However, computational overhead from ensemble voting and segmentation may constrain ultra-low-power IoT applications, suggesting optimizations like model pruning or quantization [29]. Future work could explore multi-modal fusion, combining imagery with environmental data (e.g., temperature) for improved accuracy in dynamic ocean conditions [30].

IX. CONCLUSION

The stacked CNN-ANN model introduces a novel approach to automate the classification of marine plankton species using the WHOI-Plankton 2014 dataset [19]. According to the dataset, it achieves the target success rate range by obtaining an accuracy of 81.44%. By using preprocessing techniques, the model becomes more robust. This helps give accurate results even with variable underwater conditions. While the model functions well, it also exhibits a few limitations. Using a refined subset runs the risk of missing out on rare plankton

species, which in turn limits biodiversity. Balancing computation efficiency for real-time use on low-resource devices poses another challenge [22]. Evaluating a model through only one dataset restricts the model's evaluation and generalization due to the dataset bias, like residual noise and imbalance effects. These limitations point towards a few improvement opportunities for the model. Leave for the near future to study attention mechanisms [24] and methods for enhancing computational efficiency to widen the adaptability of the model. The use of the stacked CNN-ANN alongside other data sources, for instance, environmental sensors [25], could make it useful for holistic oceanic monitoring and aid in stronger marine conservation efforts.

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